

DENVER STAPLETON, CO, USA

WMO: 724690

Lat: 39.767N

Lon: 104.869W

Elev: 5289

StdP: 12.10

Time Zone: -7.00 (NAM)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.4	5.1	-7.5	4.5	5.4	-2.4	5.9	13.8	27.2	40.8	23.8	40.8	5.0	180	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	93.9	60.7	91.2	60.0	88.5	59.6	64.5	81.8	63.4	80.7	62.3	79.7	9.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	94.7	67.2	58.5	89.1	67.0	56.8	83.9	67.4	32.3	82.0	31.3	80.3	30.5	80.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.3	19.7	17.2	-10.4	99.7	8.2	2.5	-16.3	101.5	-21.1	103.0	-25.7	104.4	-31.7	106.3		
(5)			-11.0	67.7	8.1	2.0	-16.8	69.1	-21.5	70.3	-26.0	71.4	-31.9	72.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	32.4	34.4	42.3	50.1	59.1	69.2	73.0	71.5	63.4	51.6	38.8	30.4		
	DBStd	17.55	10.51	12.19	9.56	9.38	7.41	6.60	5.07	4.90	7.37	9.29	10.57	11.59		
	HDD50	2446	546	441	268	113	16	0	0	0	6	90	357	609		
	HDD65	5667	1010	857	704	451	205	34	7	11	113	416	787	1072		
	CDD50	2971	1	4	28	116	299	576	713	666	407	140	20	1		
	CDD65	721	0	0	0	3	24	161	255	212	64	2	0	0		
	CDH74	9191	0	0	8	139	484	2066	3023	2291	988	187	5	0		
	CDH80	3800	0	0	0	19	110	921	1448	957	315	30	0	0		
	WSAvg	8.0	8.0	7.7	8.8	9.0	9.1	8.4	8.0	7.6	7.5	7.3	7.7	7.1		
	PrecAvg	15.2	0.5	0.6	1.1	1.9	2.2	1.6	1.9	1.7	1.5	1.1	0.7	0.5		
(15) Precipitation	PrecMax	21.5	1.1	1.9	3.9	5.8	5.7	3.4	4.1	4.0	7.8	2.5	2.3	2.7		
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.1	0.0	0.1	0.2	0.0		
	PrecStd	3.2	0.3	0.4	1.0	1.2	1.3	1.0	1.1	1.0	1.5	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	67.9	75.4	82.4	87.3	97.7	98.3	95.0	90.8	83.3	73.7	64.2		
		MCWB	42.7	45.6	48.1	52.9	55.3	60.0	61.3	61.1	60.0	52.7	47.3	43.4		
	2%	DB	58.5	62.1	70.5	78.3	83.0	93.0	94.4	91.8	87.2	79.3	67.1	58.4		
		MCWB	40.7	43.0	46.2	51.4	54.6	59.3	61.0	60.4	57.7	51.7	46.2	41.1		
	5%	DB	53.9	57.5	65.6	73.4	79.6	89.4	91.7	89.1	83.7	75.1	61.5	52.5		
		MCWB	38.1	40.8	44.1	50.2	54.5	59.0	60.7	60.2	56.7	50.4	43.5	38.2		
	10%	DB	49.4	52.8	60.3	68.7	75.8	85.5	88.6	85.8	80.3	70.0	55.7	47.1		
		MCWB	36.2	38.6	42.3	47.9	53.6	58.8	60.5	60.0	55.9	48.6	40.9	35.1		
	0.4%	WB	43.5	46.5	49.3	54.7	59.1	64.9	66.5	66.4	63.3	55.3	50.1	44.3		
		MCDB	61.7	66.0	73.0	76.7	75.4	82.7	86.1	80.8	84.9	73.2	67.1	61.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.8	43.3	46.6	52.5	57.4	63.3	64.5	64.4	60.4	53.0	47.1	41.4		
		MCDB	57.2	60.5	67.7	74.3	74.0	81.9	82.7	80.0	78.1	73.8	64.3	57.0		
	5%	WB	38.6	41.2	44.7	50.8	56.2	61.8	63.5	63.2	59.0	51.4	44.5	38.7		
		MCDB	53.2	56.7	63.6	71.0	73.5	80.0	81.2	79.1	76.6	71.3	60.3	52.0		
	10%	WB	36.3	38.9	42.8	48.9	54.9	60.5	62.3	62.1	57.4	49.8	41.6	35.3		
		MCDB	48.6	52.2	59.1	67.1	72.1	79.0	80.6	78.2	75.3	68.7	54.4	46.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.4	23.1	24.6	25.1	25.3	27.1	27.9	26.4	27.8	27.5	23.9	24.4		
		MCDBR	29.7	30.2	33.1	31.8	31.5	32.8	32.8	31.2	32.6	35.4	31.8	30.9		
		MCWBR	16.6	15.8	15.4	12.9	11.5	10.0	9.5	9.3	11.4	14.9	16.2	17.1		
	5% WB	MCDBR	28.9	29.4	31.3	29.8	27.4	27.4	28.1	25.9	27.8	32.2	30.5	30.4		
		MCWBR	16.5	15.7	14.9	12.5	10.9	10.3	9.2	8.9	10.8	14.1	15.7	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.271	0.302	0.320	0.328	0.358	0.348	0.310	0.273	0.254	0.243		
	taud		2.621	2.607	2.551	2.477	2.487	2.478	2.437	2.452	2.538	2.626	2.604	2.609		
	Ebn at Noon		305	316	316	309	303	299	289	290	298	302	297	296		
	Edh at Noon		21	24	29	34	34	35	36	34	29	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		818	1132	1517	1794	1983	2265	2145	1918	1662	1224	879	713		
	RadStd		35	61	115	123	156	108	95	66	88	75	50	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page